

Quantifying a Social Problem: Housing Inequality in the Netherlands

Affordability, regional disparity, and the price-to-income ratio

Group CPR02: Joey Pekel, Lior Santcroos, Benjamin Lobatto, Jeroen Warnaar, Steven van Zuijlen, Bia

15 June 2026

Contents

1	Introduction and context	1
1.1	Research questions	2
2	Data and methods	2
3	Data cleaning and merging	2
4	Visualisations	3
4.1	Temporal variation	3
4.2	Sub-population variation	5
4.3	Event association: the 2008 financial crisis	5
4.4	Spatial variation: a choropleth map	6
5	Key findings	7
6	Limitations	7
7	References	7

Public GitHub repository: <https://github.com/benjaminlobatto/woningmarkt-project> (*This link is required and must be visible in the handed-in PDF.*)

1 Introduction and context

Housing affordability is one of the most pressing social problems in the Netherlands. Over the past two decades, house prices have risen far faster than household incomes, and the gap is highly uneven across provinces: the Randstad (Noord-Holland, Zuid-Holland, Utrecht) has decoupled from peripheral regions. The OECD and UN-Habitat both treat the **price-to-income ratio (PIR)** as a core affordability indicator, with a value above 3 widely read as “unaffordable” (UN-Habitat, n.d.). The OECD’s 2025 *Economic Survey*

of the Netherlands documents a structural shortage of roughly 400,000 homes and a three-tiered market that increasingly squeezes lower- and middle-income households (OECD, 2025). De Nederlandsche Bank reaches a similar conclusion from the demand side: house prices have outpaced households’ borrowing capacity since 2013, so that by 2027 only about one in three households is projected to be able to finance an average home, down from nearly one in two in 2019 (DNB, 2024). These two independent assessments — one structural, one financial — motivate our quantitative analysis below.

This decoupling matters because it shifts wealth toward existing owners and locks younger and lower-income households out of ownership, reinforcing inter-generational inequality. We quantify the problem along three rubric dimensions — over **time**, across **provinces (space)**, and around a specific **event** (the 2008 financial crisis) — and across an income **sub-population** split.

1.1 Research questions

1. **Temporal:** How has national housing affordability (PIR) evolved over time?
2. **Spatial:** Which provinces are least affordable, and how large is the gap?
3. **Event:** How did the 2008 crisis interrupt the price trajectory?
4. **Sub-population:** Do affordability dynamics differ between higher- and lower-income provinces?

2 Data and methods

We document and merge **two public CBS datasets**, both retrieved programmatically via the `cbsodataR` OData API (collector: Statistics Netherlands; method: administrative registers and the Land Registry *Kadaster*).

Dataset	What it provides	Granularity	CBS table
House prices (existing own homes)	Average sale price, €	Year × province	83625NED
Household income	Median disposable income, €	Year × province	86161NED

We chose CBS because it is the authoritative national statistical office, its tables are openly licensed and versioned, and both indicators share a common **year × province** key that lets us merge them and compute affordability directly. The price series runs from 1995, while the regional income series begins in 2011; the merged affordability panel therefore covers **2011–2024**, and the 2008 event analysis uses the longer price series. A limitation is that purchase prices ignore the large rental/social sector (discussed in §6).

```
# Verified province-level CBS tables (confirmed via cbs_get_meta to have a
# RegioS province dimension and yearly Perioden). Each download is cached to
# data/ on first knit.
raw_price <- try_cbs("83625NED") # avg sale price, existing homes, by province
raw_income <- try_cbs("86161NED") # disposable household income, by province
```

The data source for this knit is: **Statistics Netherlands (CBS), via cbsodataR (real data)**.

3 Data cleaning and merging

`normalise_cbs()` trims CBS’ padded codes, parses the period into a numeric **year**, restricts to the 12 provinces (codes PV20–PV31), and rescales income from CBS’ “× 1000 euro” unit. We then **merge** the two datasets on **province × year** and construct **two new analytical variables**:

- **pir** — the price-to-income ratio ($\text{avg_price} / \text{median_income}$), the affordability indicator; and
- **gap** — the *affordability gap index*: PIR relative to the UN-Habitat threshold of 3 ($\text{pir}/3 - 1$), so positive values flag unaffordable province-years and the magnitude is directly interpretable.

```
panel <- inner_join(prices, incomes, by = c("province", "year")) |>
  filter(province %in% provinces) |>
  mutate(
    pir = price_to_income(avg_price, median_income), # new variable 1
    gap = pir/3 - 1 # new variable 2
  ) |>
  arrange(province, year)

# National yearly aggregates (unweighted mean across provinces).
# PIR uses the merged panel (2011-2024); the price-only series spans 1995-2024
# and is used for the 2008 event analysis.
national <- panel |>
  group_by(year) |>
  summarise(avg_price = mean(avg_price), median_income = mean(median_income),
            pir = mean(pir), .groups = "drop")
price_history <- prices |>
  group_by(year) |> summarise(avg_price = mean(avg_price), .groups = "drop")
glimpse(panel)
```

```
## Rows: 168
## Columns: 6
## $ province      <chr> "Drenthe", "Drenthe", "Drenthe", "Drenthe", "Drenthe", "~
## $ year          <int> 2011, 2012, 2013, 2014, 2015, 2016, 2017, 2018, 2019, 20~
## $ avg_price     <dbl> 203389, 193844, 181850, 185239, 194566, 205504, 213562, ~
## $ median_income <dbl> 32000, 32100, 32200, 32900, 33200, 34200, 34800, 35600, ~
## $ pir          <dbl> 6.355906, 6.038754, 5.647516, 5.630365, 5.860422, 6.0088~
## $ gap          <dbl> 1.1186354, 1.0129180, 0.8825052, 0.8767882, 0.9534739, 1~
```

4 Visualisations

4.1 Temporal variation

```
ggplot(national, aes(year, pir)) +
  geom_hline(yintercept = 3, linetype = "dashed", colour = vu_navy) +
  geom_line(colour = vu_blue, linewidth = 1) + geom_point(colour = vu_navy) +
  annotate("text", x = min(national$year), y = 3,
    label = "UN-Habitat threshold = 3", vjust = -0.6, hjust = 0,
    colour = vu_navy, size = 3) +
  labs(title = "Affordability has steadily worsened",
    subtitle = paste0("National average PIR, ", min(national$year), "-",
      max(national$year)),
    x = NULL, y = "Price-to-income ratio")
```

The PIR (years of median disposable income per average home) sits far above the UN-Habitat affordability threshold of 3 throughout and trends clearly upward, confirming a structural worsening of affordability over the observed period.

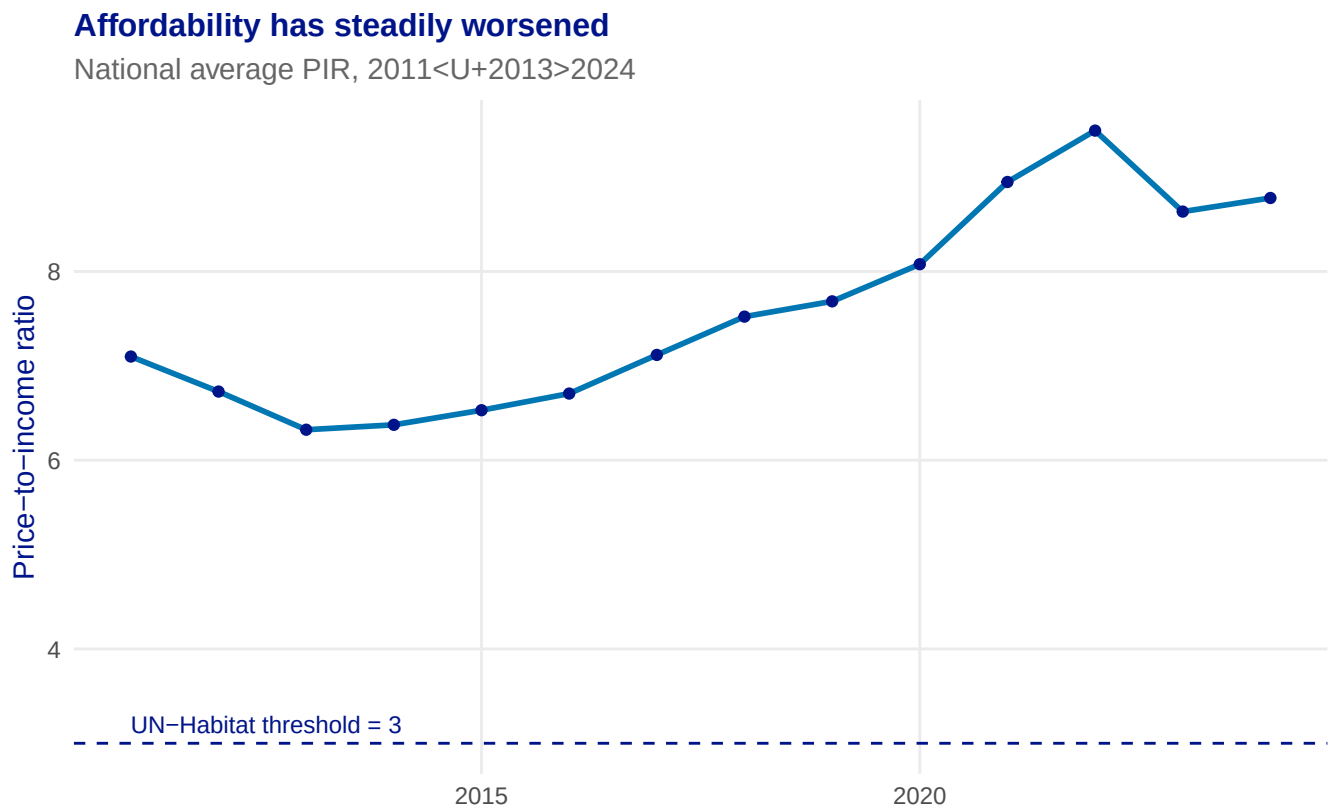


Figure 1: National price-to-income ratio over time.

4.2 Sub-population variation

```
grp <- panel |>  
  group_by(province) |> mutate(mean_inc = mean(median_income)) |> ungroup() |>  
  mutate(group = ifelse(mean_inc >= median(mean_inc),  
                        "Higher-income provinces", "Lower-income provinces"))  
ggplot(grp, aes(year, pir, group = province, colour = group)) +  
  geom_line(alpha = .5) +  
  stat_summary(aes(group = group), fun = mean, geom = "line", linewidth = 1.2) +  
  facet_wrap(~group) +  
  scale_colour_manual(values = c(vu_blue, vu_orange), guide = "none") +  
  labs(title = "Unaffordability is concentrated in richer provinces",  
       subtitle = "Thin lines = provinces; thick line = group mean",  
       x = NULL, y = "PIR")
```

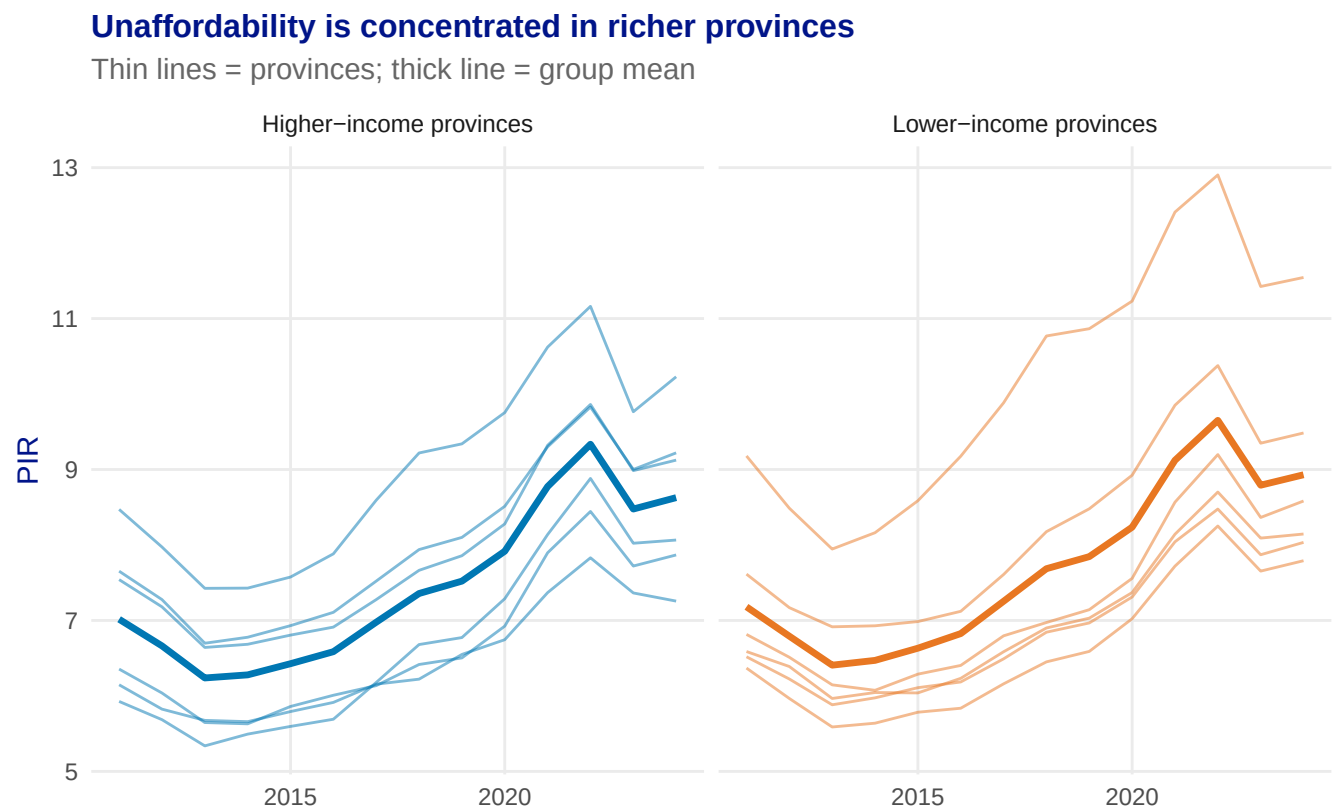


Figure 2: PIR for higher- vs. lower-income provinces.

Higher-income (largely Randstad) provinces show both higher PIR levels and faster growth, so the affordability problem is not uniform — it is regionally concentrated.

4.3 Event association: the 2008 financial crisis

```

event_df <- price_history |>
  mutate(period = ifelse(year < 2009, "Pre-2008", "2008 onward"))
ggplot(event_df, aes(year, avg_price, colour = period)) +
  geom_vline(xintercept = 2008, linetype = "dotted", colour = vu_red) +
  geom_line(aes(group = 1), linewidth = 1) + geom_point(size = 1) +
  annotate("text", x = 2008, y = max(event_df$avg_price),
    label = "2008 crisis", colour = vu_red, hjust = -0.05, size = 3) +
  scale_y_continuous(labels = label_number(prefix = "€", big.mark = ",")) +
  scale_colour_manual(values = c("Pre-2008" = vu_grey, "2008 onward" = vu_blue)) +
  labs(title = "The crisis interrupted, but did not reverse, the trend",
    x = NULL, y = "Average price", colour = NULL)

```

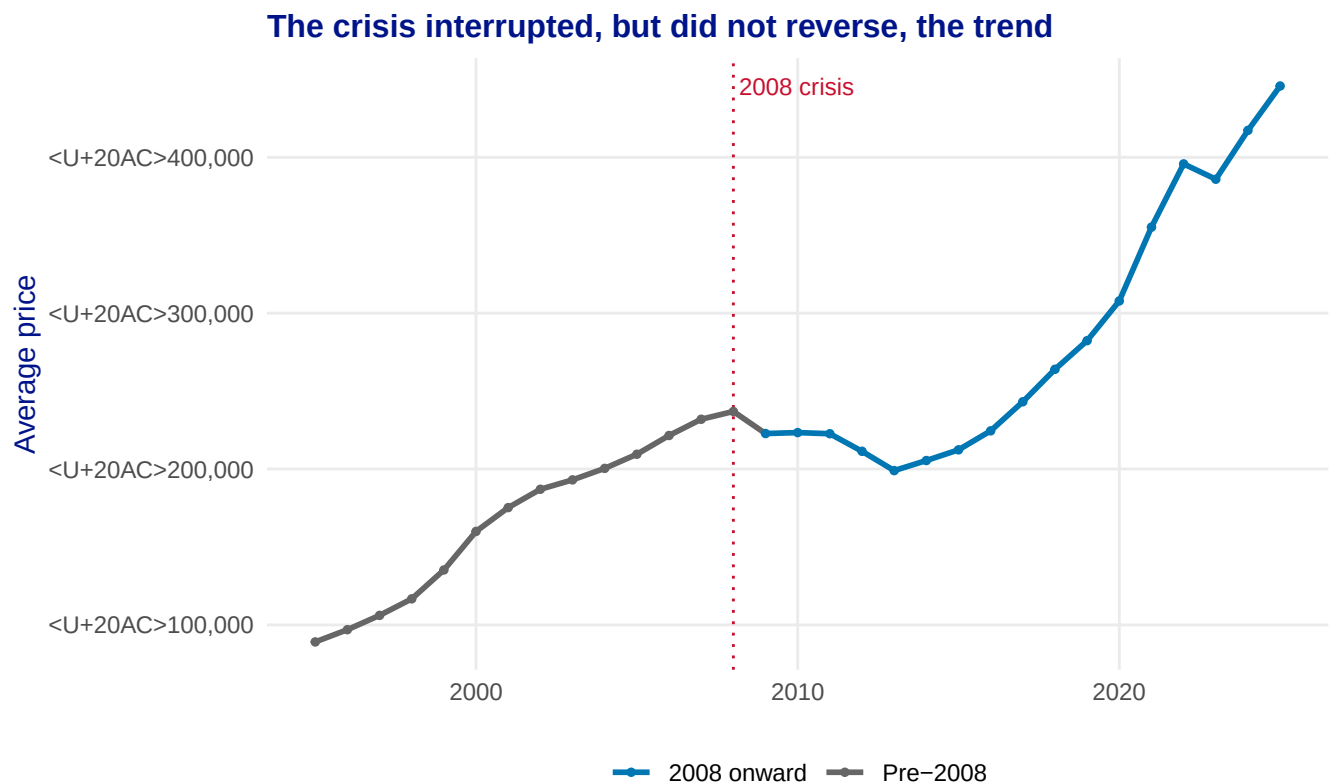


Figure 3: Average house price around the 2008 crisis (real price data, 1995-2024).

Prices fell for roughly five years after 2008 before resuming their climb, showing the crisis as a temporary correction rather than a structural break.

4.4 Spatial variation: a choropleth map

```

if (!is.null(nl) && "statcode" %in% names(nl)) {
  nl2 <- nl |> left_join(pir_latest, by = "statcode")
  ggplot(nl2) +
    geom_sf(aes(fill = pir), colour = "white", linewidth = .2) +

```

```

scale_fill_gradient(low = "#cfe8f5", high = vu_red, name = "PIR") +
labs(title = paste0("Least affordable provinces, ", latest),
     subtitle = "Darker = higher price-to-income ratio",
     caption = "Source: CBS 83625NED, 86161NED; boundaries cartomap.github.io") +
theme_void(base_size = 11) + theme(legend.position = "right")
} else {
# Fallback: ranked bar chart if the boundary file can't be fetched offline.
ggplot(pir_latest, aes(reorder(province, pir), pir, fill = pir)) +
geom_col() + coord_flip() +
scale_fill_gradient(low = "#cfe8f5", high = vu_red, guide = "none") +
labs(title = paste0("PIR by province, ", latest),
     subtitle = "Map boundaries unavailable offline - ranked bars shown",
     x = NULL, y = "PIR")
}

```

The map shows a clear Randstad-versus-periphery divide: the western provinces carry the highest price-to-income ratios, confirming that the affordability problem is spatially concentrated rather than national in scale.

5 Key findings

Nationally the PIR rose from **7.1** in 2011 to **8.8** in 2024. In 2024 the least affordable province was **Noord-Holland** (PIR 11.5) and the most affordable **Zeeland** (PIR 7.3) — a gap of **4.3** points, underscoring the regional dimension.

6 Limitations

National and provincial **averages mask within-province variation** (cities vs. rural). The analysis covers **owner-occupied purchases only**, excluding the large rental and social-housing sector that binds for many households. Figures are **nominal** unless stated, so part of the price growth reflects inflation. Because the regional income series starts in 2011, the **affordability panel spans 2011–2024**; the 2008 event analysis therefore uses the price series alone. The event analysis is descriptive and **cannot establish causality**. Finally, the PIR uses *disposable* (post-tax) income, so its levels sit above the gross-income convention behind the UN-Habitat “3” benchmark and are best read as a relative indicator across provinces and time.

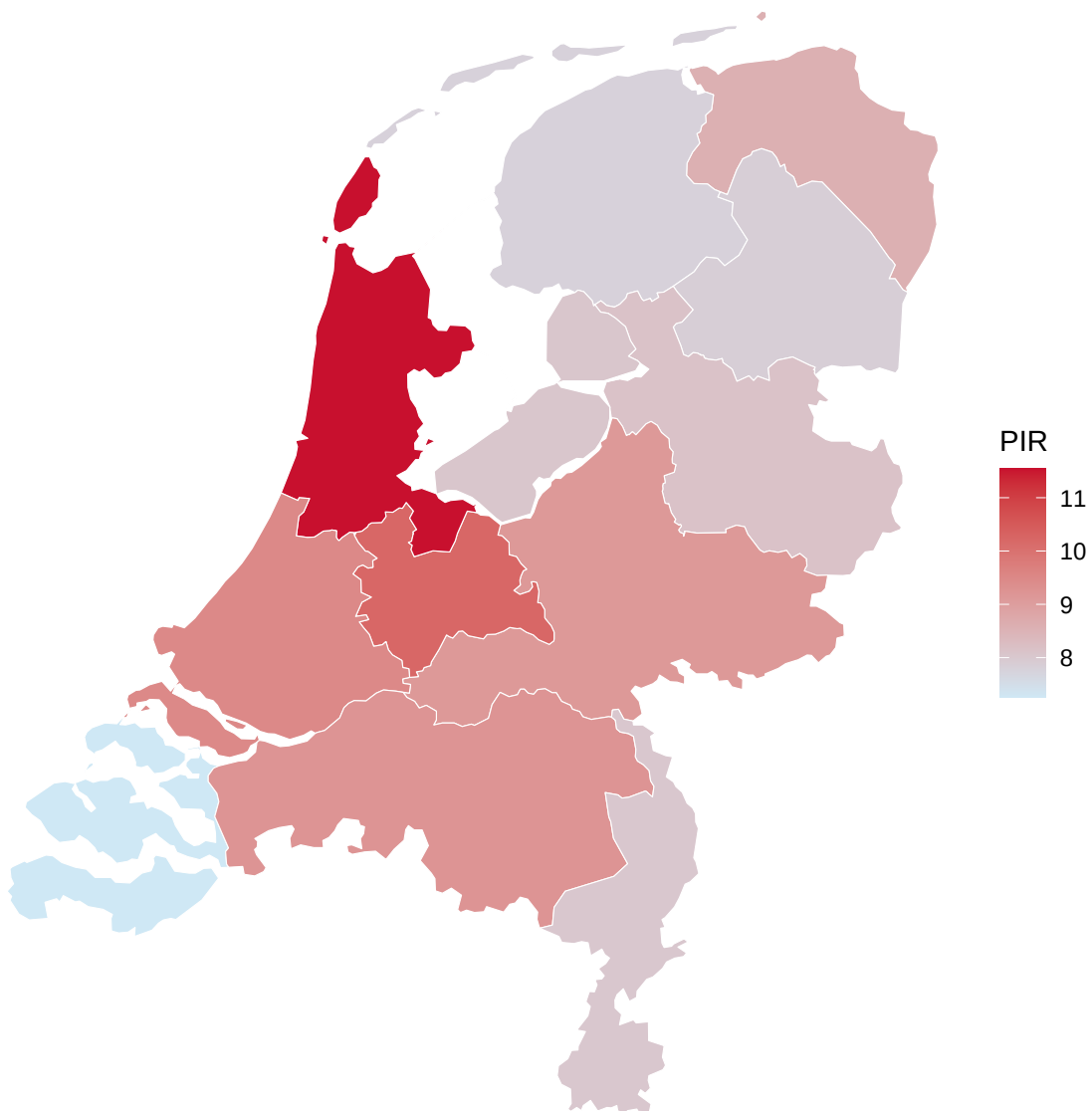
7 References

Literature (problem motivation)

- OECD (2025). *OECD Economic Surveys: Netherlands 2025 — Towards a more accessible and sustainable housing market*. OECD Publishing. https://www.oecd.org/en/publications/oecd-economic-surveys-netherlands-2025_2dd1f4aa-en.html
- De Nederlandsche Bank (2024). *Borrowing capacity and house prices* (DNB analysis). <https://www.dnb.nl/media/ccsjet4s/dnb-analysis-borrowing-capacity-and-house.pdf>
- UN-Habitat (n.d.). *Housing affordability and the price-to-income ratio*. <https://unhabitat.org>

Data sources

Least affordable provinces, 2024
Darker = higher price-to-income ratio



Source: CBS 83625NED, 86161NED; boundaries cartomap.github.io

Figure 4: Price-to-income ratio by province (latest year).

- Statistics Netherlands (CBS). *Bestaande koopwoningen; prijzen, regio* (83625NED) and *Inkomen van huishoudens; regio* (86161NED), via the `cbsodataR` package. <https://opendata.cbs.nl>
- Province boundaries: `cartomap` (<https://cartomap.github.io/nl>).

Software

- Wickham, H. et al. (2019). *Welcome to the tidyverse*. JOSS 4(43), 1686.